

Toxic Recycling: The Cost of Used Lead-Acid Battery Processing in Mexico*

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Abstract

Environmental regulation can create incentives for polluting activities to relocate across borders. Prior work has theorized and documented this pollution-haven phenomenon, but few studies measure the human capital costs borne by recipient communities. We study the January 2009 U.S. EPA tightening of the lead air-quality standard from 1.5 to 0.15 $\mu\text{g}/\text{m}^3$, which pushed used lead-acid battery recycling from the U.S. to Mexico, increasing lead pollution near Mexican recycling plants. Using nationwide ENLACE standardized tests in math and Spanish from 2006 to 2013, we compare students in schools within five miles of a plant to those farther away, before and after the policy change. Our preferred difference-in-differences estimates show reductions of 0.015 to 0.027 σ in math; Spanish effects are negative but smaller. A wind-based triple-difference isolates lead exposure from local-economy channels and yields effects of -0.07 σ in math and -0.06 σ in Spanish. Harm is concentrated in already-disadvantaged communities. Economic losses amount to at least USD 22 million per cohort exposed, in terms of foregone lifetime earnings.

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1 Introduction

Environmental regulation in high-income countries can create incentives for polluting activities to relocate across borders. Such relocation is more likely in countries with a comparative advantage in pollution-intensive production, often arising from relatively weaker regulation and enforcement.¹ Significant efforts have been put towards documenting *when* this shifting of pollution may take place (e.g., [Copeland and Taylor \(1994\)](#) and [Copeland et al. \(2022\)](#)), but there has been little focus on the *effects* of this relocation on recipient communities. Among pollutants, lead is especially damaging in this context: it is widely used in industry, has no safe level of exposure, and even small amounts can permanently impair cognitive development ([Centers for Disease Control and Prevention, 2013](#); [Clay, Hollingsworth, & Severnini, 2024](#)).² Children are particularly vulnerable to cognitive and other health effects because they absorb more lead and suffer greater developmental harm ([Rees & Fuller, 2020](#)).

We study the effects of one such case — the relocation of one of the most lead-intensive industrial activities worldwide, used lead-acid battery (ULAB) recycling, from the U.S. to Mexico. We estimate some of the costs of this pollution relocation, specifically those related to educational achievement. In 2009, the U.S. Environmental Protection Agency tightened regulatory standards for lead, successfully reducing airborne lead concentrations around battery recycling plants in the U.S. and prompting ULAB recycling activities to shift to Mexico ([Tanaka, Teshima, & Verhoogen, 2022](#)).

To identify causal effects in Mexico, we use a difference-in-differences design, comparing children attending schools close to and farther away from battery recycling sites, before and

¹There is no consensus on whether stricter environmental regulations in rich countries systematically displace dirty industries to poorer ones: some reviews find evidence consistent with pollution haven effects ([Copeland, Shapiro, & Scott Taylor, 2022](#)), while others argue that broad displacement is limited, aside from rare case studies ([Levinson, 2023](#)).

²Lead is a neurotoxin: it affects cardiovascular and renal health and, in some cases, exposure can be lethal ([Ara & Usmani, 2015](#)).

after the 2009 U.S. regulatory change that shifted battery recycling activities to Mexico.³ We have data on the locations of the 26 authorized battery recycling facilities and of all schools in Mexico. We measure academic achievement using student-level panel data from a nationwide standardized exam that tested math and Spanish proficiency between 2006 and 2013. We observe the universe of students tested in this exam, which, at its peak, administered tests to almost 14 million students each year.

Exposure to lead-battery recycling reduces academic achievement. The post-2009 increase in recycling activity negatively affects students' learning in schools within five miles of a battery recycling plant. In our preferred specification, which includes school-session-grade fixed effects and plant-by-year fixed effects, math test scores in these schools decreased between 0.015 and 0.027 standard deviations (σ) after 2009, relative to students in schools farther away but in the same state. For Spanish, we also estimate negative effects, though slightly smaller (~ 0.007 - 0.019σ). The pre-2009 trends in test scores do not differ by distance to a recycling plant (supporting the identification assumption of the difference-in-differences design). Further, the event studies suggest that the effects of the increase in lead pollution worsen over time.

The results are robust to alternative specifications that guard against other determinants of learning that may also vary with distance to the recycling plant. For example, factors such as growth in labor demand, which could have accompanied the post-2009 increase in recycling activity, could affect wages and local economic activity near the plants. This, in turn, could affect household incomes and children's education. We can test this by defining exposed schools not only by distance but also by prevailing wind direction. The addition of wind direction helps us more precisely identify lead-exposed students, as environmental studies of lead pollution around battery recycling plants and other lead smelters show that pollution levels correlate with prevailing winds (Ettler, 2016). Using

³Mexico lacks a nationwide lead monitoring network, so we cannot directly track exposure levels. Instead, we estimate the reduced-form effects of the 2009 policy shift on children's outcomes near recycling plants in Mexico.

a triple-difference estimate, comparing test scores at schools nearby *and* downwind from those at (i) nearby but upwind schools, (ii) schools farther away in the same state, and (iii) the corresponding pre-2009 periods, we find larger effect sizes (-0.07σ for math and -0.06σ for Spanish, both significant at the 1% level).

Another concern is selective sorting, driven by both the post-2009 increase in pollution and the potential economic changes associated with the increase in battery recycling. An increase in battery recycling could induce migration if families either move away from the area due to increased pollution or move toward it because of new economic opportunities. Non-random migration could bias our results. Consistent with this concern, schools near recycling plants exhibit modestly higher enrollment levels after 2009 than those farther away. However, several robustness checks mitigate this concern. First, assigning exposure based on each student’s 2008 school location (analogous to an “intent-to-treat” design) yields estimates of -0.016σ for math and -0.014σ for Spanish. Second, restricting the sample to students whose “Near status” does not change over the panel yields estimates of -0.026σ for math and -0.019σ for Spanish. Third, a specification that includes student fixed effects absorbs any time-invariant differences between students who sort in or out, and the estimated effects remain negative and statistically significant. Finally, the triple-difference estimates cited above, comparing students in downwind to upwind schools, are robust to any sorting that does not itself correlate with prevailing winds.

Finally, we examine whether and how lead exposure affects different individuals and communities. A key strength of our data and setting is that we observe changes in lead-emitting industrial activity across many locations, in contrast to studies that focus on a single polluted site (Rau, Urzúa, & Reyes, 2015). This enables us to examine how effects differ across different communities. We use census data to characterize baseline socioeconomic characteristics of the communities in which schools are located. Across multiple measures of socioeconomic welfare, the negative effects of the post-2009 increase in battery recycling are stronger in worse-off communities. For example,

we can compare the effects in communities with different levels of marginalization, as defined by the Mexican government as a composite measure of literacy rates, educational attainment, housing quality, infrastructure access, and income. In more marginalized communities, with above-median marginalization index values, we estimate a -0.052σ effect on math scores (p-value < 0.001), compared to a small, statistically insignificant positive effect (0.003σ) in below-median communities. We find qualitatively similar results when comparing effects across communities with adult education levels above and below the median, infrastructure quality, formal employment rates, income inequality, and malnutrition rates. These results suggest that lead exposure may exacerbate inequality in educational achievement.

What can be done, from a policy perspective, to address these effects? One option is to increase regulatory stringency and/or enforcement to reduce lead emissions from battery recycling facilities. In section 5, we estimate the cost of the education effects we identify here. In terms of lost lifetime earnings for exposed students, the costs are on the order of 22.3–55.9 million USD per cohort exposed. This estimate ignores the other health costs associated with lead pollution and likely outweighs the expected costs of lead pollution control technology.

This study offers three main contributions to the literature on the effects of lead pollution exposure. First, our findings contribute to the pollution haven and trade literature, which examines shifts of pollution-intensive activities toward countries with weaker environmental regulation (World Bank, 2019; Copeland et al., 2022; Copeland & Taylor, 2004). Much of this literature has focused on trade flows and firm location decisions (Brunnermeier & Levinson, 2004), the effects of these shifts on emissions (Shapiro, 2021), the emission content of trade (Wang & Yang, 2020; Bombardini & Li, 2020), and the effects of environmental regulation on the competitiveness of pollution-heavy industries (Levinson & Taylor, 2008; Broner, Bustos, & Carvalho, 2012; Cherniwchan & Najjar, 2022). By contrast, we provide direct causal estimates of the welfare costs in a middle-income

country resulting from regulatory changes in a high-income country.⁴ Our results complement evidence showing that the distributional consequences of trade and foreign investment extend beyond wages and employment to encompass health and human capital. Although increased production may also raise local incomes and, in theory, improve educational outcomes, our findings suggest that in this context the adverse effects of pollution on human capital dominate. This suggests that the true costs of pollution have exceeded what is typically measured in terms of production-side outcomes, as they include persistent human capital losses for affected populations.

Second, while previous research has estimated the causal effects of lead exposure on learning, IQ, and other educational outcomes in high-income countries such as the United States (Sorensen, Fox, Jung, & Martin, 2019) and Sweden (Gronqvist, Nilsson, & Robling, 2020), few papers have assessed these effects in low- and middle-income countries.⁵ A broader, recent literature on air pollution and learning *has* expanded into middle-income contexts, including Tehran (Heyes & Saberian, 2024), China (Deng, Du, Guo, Sun, & Xia, 2023), and Bogotá (Villalobos & Blackman, 2025), and consistently finds that pollution exposure during schooling or exams reduces achievement. Villalobos and Blackman (2025) further document, through a meta-analysis of comparable studies, that these air pollution effects are systematically larger in middle-income than high-income settings.⁶ Thus, extrapolating from high-income settings likely underestimates the true costs of lead pollution exposure in poorer contexts, where emissions are higher, institutions are weaker, and households are less able to offset damage through educational investments, medical care, or pollution-avoidance behaviors (Arceo, Hanna, & Oliva,

⁴Two closely related papers are Tanaka et al. (2022), who study the impact of the same regulatory change in the U.S. as us on maternal and infant health outcomes in Mexico; and Bombardini and Li (2020) who show export booms raise SO_2 , and as a result, infant mortality in China.

⁵Two notable exceptions are: Rau et al. (2015), who use data from one single site in the city of Arica in northern Chile to study the relationship between school proximity to toxic lead waste and students' scores in a standardized test for fourth graders and a college entrance exam; and Tanaka et al. (2022), who study the effects of ULAB recycling activity in Mexico on maternal and infant health outcomes.

⁶Heterogeneity analyses from recent studies in the U.S. additionally suggest that the effects of pollution are worse in poorer or otherwise disadvantaged communities, both for lead exposure (Hollingsworth, Huang, Rudik, & Sanders, 2025) and for fine-particulate exposure (Gilraine & Zheng, 2024; Inafuku, Halliday, Lusher, & de Paula, 2026).

2016). Our study contributes to this literature by providing direct causal estimates of one lead-pollution-exposure channel in a middle-income country, complementing recent meta-analytic global estimates that apply rich-country dose-response relationships to low- and middle-income country (LMIC) blood-lead distributions (Crawford, Todd, Hares, Sandefur, & Silverman Bonnifield, 2025). Further, consistent with previous results, our heterogeneity analysis suggests that, even within Mexico, effects are concentrated in more marginalized communities. This supports the evidence that underlying levels of development matter when estimating pollution costs and that extrapolations from rich-country estimates may understate the true costs in less developed settings.

Finally, from a methodological perspective, previous studies have primarily relied on cross-sectional data from a single grade or educational data aggregated at the grade or school level to study the effects of lead exposure on educational outcomes (Rau et al., 2015; Aizer, Currie, Simon, & Vivier, 2018; C. Persico, Figlio, & Roth, 2020; Zheng, 2021; Jacqz, 2022; Sauve-Syed, 2024).⁷ In contrast, our paper tracks individual students' academic performance over several years to estimate the impact of lead pollution on cognitive development, as proxied by math and Spanish test scores. Hence, the longitudinal structure of our data allows us to address student sorting as a confounding factor, a common challenge in the existing literature (Rau et al., 2015). We also expand on this literature by focusing on test scores for grades 3-6 and grade 9 from 2006 to 2013, providing a more robust measure of cognitive development by capturing changes in individual students' academic achievement over time.

⁷Three notable exceptions are Gronqvist et al. (2020), Hollingsworth et al. (2025), and Gazze, Persico, and Spirovska (2024), who also exploit longitudinal data. However, all three focus on high-income country contexts (Sweden and the United States), and Gazze et al. (2024) estimate longer-run peer-spillover effects of lead-exposed classmates rather than the direct effect of own exposure.

2 Relevant Background Information

2.1 Lead-acid battery recycling

The production of lead-acid batteries accounts for over 86 percent of the total global demand for lead ([International Lead and Zinc Study Group, 2023](#)), reflecting a growing market driven by rising vehicle ownership in low- and middle-income countries and global demand for energy storage solutions ([Rees & Fuller, 2020](#)). At the same time, lead demand is primarily met by secondary lead, i.e., lead from recycling, rather than by primary lead that is mined. Used lead-acid batteries (ULABs) have one of the highest end-of-life recycling rates ([Alistair, Binks, & Gediga, 2016](#)), and almost all of the lead in them can be reclaimed. Furthermore, recycling lead from used lead-acid batteries is a cost-effective alternative to mining, contributing to high recycling rates. Still, recycling lead is dangerous and can be heavily polluting if strict protocols and advanced technology are not used to control lead emissions during the process ([Ballantyne, Hallett, Riley, Shah, & Payne, 2018](#)). Lead from recycling can enter the environment through several pathways. Lead-laden gases and dust can escape during handling, separation, smelting, and refining ([U.S. Environmental Protection Agency, 1995](#)). Lead can also run into the environment through water discharges or the dumping of solid wastes. Moreover, where worker safety protocols are non-existent or not enforced, workers can expose themselves and their families to lead contamination by bringing lead dust home on their clothes and skin ([WHO, 2017](#)). Thus, improper management of even small-scale lead recycling operations can result in environmental contamination and pose a severe threat to human health ([Commission for Environmental Cooperation, 2013](#); [WHO, 2017](#)).

A growing literature from environmental science and public health scholars documents that ULAB recycling is among the largest sources of population lead exposure in LMICs. [Ericson et al. \(2016\)](#) estimate between 10,599 and 29,241 informal ULAB processing sites operate across 90 LMICs, exposing 6 to 16.8 million people and producing geometric

mean blood lead levels of 31 $\mu\text{g}/\text{dL}$ among children and 21 $\mu\text{g}/\text{dL}$ among adults living near these sites.⁸ Modeling that allows exposure to extend beyond the immediate vicinity of recycling sites suggests ULAB recycling may account for around a third of the total LMIC lead burden, with a 95 percent confidence interval whose lower bound exceeds 20 percent (Crawford, Hu, & Mitchell, 2026). Field studies corroborate these global findings. Surface-soil and airborne lead concentrations near ULAB and lead-smelting sites typically exceed background levels by an order of magnitude and decline with distance from the facility, with the spatial gradient tied to prevailing wind direction (Adventini, Santoso, Lestiani, Syahfitri, & Rixson, 2017; Benin, Sargent, Dalton, & Roda, 1999). A review of biomonitoring studies near lead-acid battery manufacturing and recycling plants in developing countries finds that the average child living near such a facility has a blood lead level of about 29 $\mu\text{g}/\text{dL}$ (Gottesfeld & Pokhrel, 2011). Even in more regulated settings, exposure persists: Johnston, Franklin, Roh, Austin, and Arora (2019) detect pre- and post-natal lead in shed baby teeth from children who lived within two miles of a secondary lead smelter in Los Angeles, California during its years of operation. Closer to our setting, Garcia-Vargas et al. (2014) document spatial clustering of elevated blood lead in adolescents aged 12 to 15 living adjacent to and downwind of the Met-Mex Peñoles lead-zinc smelter in Torreón, Coahuila, the world’s fourth-largest lead-zinc smelter, with stronger clustering on the side of the prevailing wind. Although Peñoles is a primary smelter rather than a ULAB recycler, the spatial pattern they document is the same one we exploit in our wind-based identification strategy in Section 4.1.

In our study, we focus on used lead-acid battery recycling in Mexico and study the period between 2006 and 2013. The battery recycling industry in Mexico comprises both formal and informal firms. Following Tanaka et al. (2022), we use a list of 26 authorized battery-recycling plant locations in Mexico, compiled from the Commission for Environmental Cooperation (2013), RETC records, and a SEMARNAT list of approved

⁸For comparison, the U.S. Centers for Disease Control and Prevention’s blood lead reference value for children is 3.5 $\mu\text{g}/\text{dL}$, although the CDC emphasizes that no level of lead in blood is known to be safe (Centers for Disease Control and Prevention, 2022, 2013).

battery recyclers. This list identifies authorized facility locations used to define exposure (see Figure 1 for the locations of these facilities) (Tanaka et al., 2022; Commission for Environmental Cooperation, 2013).⁹ The Commission for Environmental Cooperation (CEC) estimated that Mexico may have been producing about 401,151 metric tons of secondary lead annually around this time (Commission for Environmental Cooperation, 2013). In Mexico, the Secretary of the Environment and Natural Resources (*Secretaría de Medio Ambiente y Recursos Naturales* or SEMARNAT) and the Secretary of Health (*Secretaría de Salud*) regulate airborne lead concentrations; the standard during our study period was 1.5 micrograms per cubic meter.¹⁰ Companies are also required to report emissions of toxic materials via the Registry of Emissions and Transfers of Contaminants (*Registro de Emisiones y Transferencia de Contaminantes* or RETC) program. Enforcement of these regulations is incomplete and inconsistent (Gottesfeld, Chavez Arce, & Macias Raya, 2023). More than half of the registered recycling facilities in Mexico did not report emissions during our study period, and air quality monitoring outside of Mexico City was almost non-existent (Commission for Environmental Cooperation, 2013). During our study period, the country lacked standards around the construction and operation of battery recycling plants, including emissions from stacks (Commission for Environmental Cooperation, 2013).¹¹ These regulatory gaps compound the problem: lead-abatement technology is costly, and absent a regulatory mandate, Mexican recycling plants have little incentive to adopt it. Consequently, based on reported emissions data from RETC and TRI, OK International and Fronteras Comunes (2011) estimate that air lead emissions from Mexican battery-recycling plants were approximately 20 times higher than emissions from U.S. plants of similar permitted capacity.

⁹In addition to authorized facilities, an unknown number of small, unregulated secondary lead smelters may have operated in Mexico during this period, though Commission for Environmental Cooperation (2013) could not document the extent of this activity.

¹⁰The current standard is 0.50 micrograms per cubic meter under standard NOM-026-SSA1-2021.

¹¹During this period, Mexico's occupational airborne lead exposure limit was also three times the U.S. limit, 150 µg/m³ compared with 50 µg/m³ (OK International & Fronteras Comunes, 2011).

In January 2009, the U.S. Environmental Protection Agency reduced the air quality standard for lead from 1.5 $\mu\text{g}/\text{m}^3$ to 0.15 $\mu\text{g}/\text{m}^3$.¹² This regulation made it more difficult and expensive to recycle used lead-acid batteries domestically, pushing U.S. recycling facilities to close or shift activities to Mexico. Tanaka et al. (2022) document this shift, showing that the U.S. regulatory change drastically increased used battery exports from the U.S. to Mexico and increased the value of the lead battery recycling industry in Mexico, relative to other, similar industrial sectors. It is estimated that, as of 2013, 30-60 percent of all used batteries processed in Mexico came from the U.S. (Commission for Environmental Cooperation, 2013).

2.2 Health effects of lead pollution exposure

Lead poses a toxic threat regardless of how it enters the body. The U.S. Centers for Disease Control and Prevention emphasizes that no level of lead in blood is known to be safe, so any exposure is hazardous, especially for fetuses, infants, and young children (Centers for Disease Control and Prevention, 2022, 2013). At high levels, lead can be lethal (WHO, 2017). At low levels, it acts as a cumulative toxicant that damages multiple organ systems. The developing nervous system is most vulnerable: the prefrontal cortex of children is particularly sensitive to lead exposure, so early-life exposure can permanently impair cognition, behavior, and learning (WHO, 2017; Centers for Disease Control and Prevention, 2013).

Empirical evidence from high-income countries confirms these mechanisms. Childhood lead exposure lowers IQ (Aizer et al., 2018; Gronqvist et al., 2020), reduces test scores and educational attainment (Reyes, 2015; Aizer & Currie, 2019; Hollingsworth et al., 2025; Sauve-Syed, 2024), and raises later-life antisocial behaviors, including arrests and incarceration (Ferrie, Rolf, & Troesken, 2012; Billings & Schnepel, 2018; Sorensen et al., 2019; C. Persico et al., 2020; C. Persico, 2026). Recent work documents that the effects of lead are larger when exposure spans more years of childhood (Hollingsworth

¹²While signed in May 2008, it became effective in January 2009.

et al., 2025) and that they spill over from lead-exposed children onto unexposed classmates, lowering high-school graduation and SAT-taking rates for the broader cohort (Gazze et al., 2024). Further, Curci and Masera (2025) find that lead poisoning raised urban crime enough to drive suburbanization.

Regulatory measures such as removing lead from gasoline, banning lead in paint, and setting ambient air-quality standards have substantially reduced exposure in rich countries. For example, the U.S. reduced airborne lead pollution by about 90 percent between 1980 and 2016 (U.S. Environmental Protection Agency, 2014) and continues to tighten regulation of remaining sources. Industrial lead emissions into air, water, and soil nevertheless remain a major exposure pathway in both developed and developing countries, with emissions estimated to be increasing in China and Mexico (Clay et al., 2024).

2.3 The Mexican Education System

The Mexican education system is based on a federal model, regulated by the Ministry of Education (*Secretaría de Educación Pública* or SEP), while individual states are responsible for administration and implementation. School days are shorter than in many other countries, with two sessions each day to accommodate all students. These are divided into morning (8:00 am to 12:30 pm) and afternoon (2:00 pm to 6:30 pm) sessions. While these two sessions take place within the same physical school, there are essentially two different operations, with different teachers and principals.¹³

The system is divided into three main levels: Basic, Upper Secondary, and Higher Education. Basic education, which includes preschool, primary, and lower secondary education, is compulsory and free for all students. Preschool lasts three years, primary lasts six years (grades 1-6), and lower secondary lasts from grades 7 to 9. Upper secondary education (grades 10–12) provides students with specialized academic or

¹³Starting in 2007, Mexico began rolling out the Full-time Schools Program (*Programa Escuelas de Tiempo Completo* or PETC), which increased the school day from 4.5 to 6-8 hours (Padilla-Romo, 2022; Cabrera-Hernández, Padilla-Romo, & Peluffo, 2023). The program eventually reached more than 25,000 schools before being terminated in 2022. We do not expect this program to correlate with battery recycling plant locations, so it is not a threat to our causal estimates.

vocational training. It was not compulsory for most of our study period, but it is available to all students who have completed lower secondary education. Upper secondary education became compulsory in 2012 ([Santiago, McGregor, Nusche, Ravela, & Toledo, 2012](#)), but parents were responsible for ensuring their children complete grade 12, and enforcement varies across states.

Public schools account for about 89 percent of the enrollment.¹⁴ By 2013, more than 25 million students were enrolled in over 243,000 schools. This included about 4 million in preschool, 14.1 million in primary education, and 7.3 million in lower secondary education.

The Mexican education system has undergone significant reforms over the past two decades, with a focus on improving the quality of teaching and learning, as well as increasing access to education. These reforms have included changes to teacher training and evaluation, the introduction of new curricula and teaching materials, and the development of standardized tests to assess student learning, such as the National Assessment of Academic Achievement in Schools (ENLACE), whose data we use.

3 Data

For our analyses, we use the following data sources.

3.1 Used Lead-Acid Battery Recycling (ULABs) Facilities

To identify the effects of lead exposure from battery recycling, we use a list of 26 used lead-acid battery recycling facilities in Mexico (see [Figure 1](#) for plant locations). These facilities were identified and geo-referenced by [Tanaka et al. \(2022\)](#). The authors identified facilities from three main sources: (1) a report published by the Commission for Environmental Cooperation ([Commission for Environmental Cooperation, 2013](#)); (2) from Mexico's Registry of Emissions and Transfer of Contaminants, under which plants

¹⁴When attending public schools, students usually enroll in the closest one to their homes. For new students, their parents submit a list of preferred schools. Assignment is carried out considering criteria such as: 1) siblings attending the school and 2) household address (proximity), among others, established by each State.

in Mexican industries that emit select pollutants are required to register;¹⁵ and (3) from a 2014 list of approved battery recyclers from Mexico's Secretariat of the Environment and Natural Resources. Given this search method, we include only authorized battery-recycling plants in the sample, and our analyses will only address lead pollution from these sources and ignore any lead pollution emitted from informal battery recyclers.¹⁶

The most comprehensive facility roster is drawn from a 2014 SEMARNAT list, which postdates the student data used in this study (2006–2013). These plants held long-standing permits and are documented to have operated through the late 2000s and early 2010s. We are not aware of any significant post-2006 entrants, and any facilities incorrectly included that opened after our study window would bias our estimates downward, thereby attenuating effects toward zero.

3.2 Wind direction

We accessed wind direction data from the Global Wind Atlas.¹⁷ The Global Wind Atlas reports microscale modeled wind climate data at a 3-kilometer resolution grid. For each grid cell, the atlas reports average wind speed and direction. For each of the 26 battery recycling facilities listed above, we extracted the 100-meter wind frequency rose data for the grid cell containing the facility. These wind frequency roses show the frequency with which the wind blows from a given direction at 100 meters above ground level. The 360 degrees around a point are divided into 12 30-degree sectors, and the fraction of time the wind blows in each sector is reported.

¹⁵Unfortunately, the reported emissions from Registry of Emissions and Transfer of Contaminants are an unreliable measure of lead emissions from ULAB facilities in Mexico. Environmental organizations have reported that many large recycling facilities do not report to the Registry of Emissions and Transfer of Contaminants [Gottesfeld et al. \(2023\)](#). Indeed, our data confirms that many facilities do not report to the registry and, for those that do, the reported emissions often contain irreconcilable errors.

¹⁶In other countries, informal recycling, done without regulatory oversight and often located in residential areas, is common and can drastically increase lead exposure ([Haeffliger et al., 2009](#); [Ericson et al., 2016](#); [Kinally, Antonanzas-Torres, Podd, & Gallego-Schmid, 2024](#)).

¹⁷Global Wind Atlas 3.0 is a free, web-based application developed, owned, and operated by the Technical University of Denmark (DTU). The Global Wind Atlas 3.0 was released in partnership with the World Bank Group, using data from Vortex and funded by the Energy Sector Management Assistance Program (ESMAP). For additional information visit <https://globalwindatlas.info>.

3.3 Test Scores from the National Assessment of Academic Achievement in Schools (ENLACE)

Our primary outcomes of interest, math and Spanish test scores, come from a national standardized test, the National Assessment of Academic Achievement in Schools (*Evaluación Nacional del Logro Académico en Centros Escolares* or ENLACE). This nationwide test was designed to evaluate students' academic performance in public and private schools. It was a low-stakes test for students, as it did not influence their GPA, graduation, or admissions to any school. The exam was developed and administered by the Ministry of Education (*Secretaría de Educación Pública* or SEP), with oversight from the National Institute of Education Evaluation (*Instituto Nacional para la Evaluación de la Educación* or INEE). It aimed to provide information on the quality of education in Mexico, to measure the effectiveness of educational policies and practices, and to offer parents and educators information on students' and schools' performance.

ENLACE was implemented for the first time in 2006 and was administered annually until 2013 to all public and private elementary school students in grades 3 to 6, as well as to students in grade 9.¹⁸ From 2009 to 2013, it was also administered to students in grades 7 and 8, and from 2008 to 2013 to students in grade 12. The exam tested students in Spanish, mathematics, and a rotating subject. It consisted of 50 to 70 multiple-choice questions per subject applied in 45-minute sessions over two days. Questions were designed to assess different levels of cognitive complexity, ranging from information recall to analysis and synthesis. The exam was usually administered in April, and the results were disseminated to parents, educators, and schools in August of the same year.

The test was designed as follows. To ensure comparability of scores across exam versions, overlapping questions are used across years, and SEP uses a calibration process called Test Equating based on Item Response Theory, which determines scores based on the number

¹⁸The exam was discontinued in 2013 as part of the education reform undertaken by the 2012–18 government.

of correct answers and which questions were answered correctly.¹⁹ After calibration, the students' scores were standardized on a scale of 200 to 800, with a mean of 500 and a standard deviation of 100 for each subject and grade. This enabled an accurate comparison of scores across years for each subject and grade.

We observe math and Spanish scores for 30,674,951 individual students in 151,327 schools across the country from 2006 to 2013 (91,005,860 observations in total). These data correspond to the universe (after removing inconsistent entries) of students who sat for the ENLACE test during this time. These data, except for a few cases where the exam could not be held for various reasons, pertain to all students in Mexico who were eligible for the test in a given year.²⁰

In addition to the test scores, the dataset contains for each student-year observation:

- An identifier of the school in which the student was enrolled and took the test;
- A student identifier that is a scrambled version of the student's national identification number, the Unique Population Registry Code (*Clave Única de Registro de Población* or CURP), so we are able to track individual students over time;
- An indicator of the grade in which the student was enrolled at the time of the test; and
- An indicator of the school session, AM or PM, in which the student was enrolled at the time of the test.

¹⁹Test equating is a statistical method that adjusts for differences in the difficulty level of different versions of each exam so that scores are comparable across years. The model assesses levels of difficulty, guessing, and discrimination and assigns a score to each student.

²⁰In 2007, the exam reached 98% of schools nationwide (Avilés, 2007). Yet, Oaxaca state is largely missing from the data. In 2012, teacher unions prevented the test from being administered altogether (Notimex, 2012), and in 2011 it was limited to students from the National Council for Educational Development (*Consejo Nacional de Fomento Educativo*, CONAFE), covering only 13% of schools and 0.9% of the state's student population. The other notable state is Michoacán, where coverage declined from 96% of students in 2007 to 42% in 2011 (SEP, 2011).

3.4 Location of schools

We retrieved the location of all schools in Mexico from the National Institute of Education Evaluation (*Instituto Nacional para la Evaluación de la Educación* or INEE).²¹ Specifically, we downloaded the Comprehensive Assessment Results System (*Sistema Integral de Resultados de las Evaluaciones* or SIRE) database for each state and extracted the GPS coordinates for each school in the database. This database was last updated for the 2016-17 school year; therefore, it may not include schools that were operating during the ENLACE testing period but have since closed. However, we can match 90.9% of the schools in the test-score data and 93.7% of the students.²² The resulting data set has a total of 84,336,559 observations (28,730,665 individual students in 137,529 schools).

3.5 Locality and municipality characteristics

For the heterogeneity analysis described in section 4.3.3, we use locality and municipality characteristics from various sources. A locality in Mexico is the smallest officially recognized territorial unit used in population and geographic statistics. Municipalities are larger administrative units between localities and states. Based on the school location data described above, we can associate each school with its corresponding locality and municipality. We obtain locality- and municipality-level demographic and socioeconomic data from various statistical archives.²³ The variables we use include adult education levels (years of completed education for individuals older than 15), access to infrastructure (percent of households in locality with access to sewerage), rates of social security

²¹We downloaded the SIRE database for each state from this url: <https://www.inee.edu.mx/evaluaciones/sire/sire-bases-de-datos/>. We last downloaded it in July 2023.

²²We compared observable characteristics (e.g., enrollment size, average test scores) of matched versus unmatched schools and found no meaningful differences, suggesting that unmatched schools are unlikely to bias the estimates.

²³Locality-level measures come from a 2005 population enumeration from Mexico's National Population Council (*Consejo Nacional de Población* or CONAPO) ([Instituto Nacional de Estadística y Geografía, 2005](#)). Municipality characteristics come from the National Council for Evaluation of Social Development Policy (*Consejo Nacional de Evaluación de la Política de Desarrollo Social* or CONEVAL) and the National Population Council (*Consejo Nacional de Población* or CONAPO) ([Consejo Nacional de Población \(CONAPO\), 2005](#); [Consejo Nacional de Evaluación de la Política de Desarrollo Social \(CONEVAL\), 2005](#)).

access, malnutrition rates, local income inequality (Gini index), and the marginalization index (a proxy means test of poverty at the locality level).²⁴

4 Estimating the Effects of Lead Exposure on Test Scores

In this section, we explain how we estimate the average impact of lead-acid battery recycling on students' academic performance. Tanaka et al. (2022) showed that after the US EPA strengthened its air quality standard for lead in 2009, used lead-acid battery exports from the US to Mexico sharply increased relative to previous years. The value added and output in Mexican battery-recycling plants also increased discontinuously relative to those of similar industries. We investigate the potential effects of this increased battery recycling activity on academic performance in schools located near these recycling plants, where students can be directly exposed to lead contamination primarily through the air and topsoil.

4.1 Defining lead-exposed students

The gold standard for measuring lead pollution exposure is a biological measure of lead in the body. This can be done by testing blood to measure lead concentrations from recent exposure (e.g., Gazze et al. (2024)), or testing bones or teeth (e.g., shed baby teeth à la Johnston et al. (2019)), to measure longer-term lead exposure. These tests provide individual measures of lead exposure. When these measures are unavailable, researchers can use environmental pollution data as a proxy for lead exposure (e.g., Hollingsworth and Rudik (2021)). Lead in the air, soil, or water is a likely indicator that populations living, studying, or working in the polluted area have been exposed. Unfortunately, Mexico lacked a lead pollution monitoring network, and it did not systematically collect biomarkers of lead exposure during our study period.²⁵ To overcome this, we

²⁴The marginalization index is calculated by Mexico's *Consejo Nacional de Población*, or the National Population Council, and uses principal component analysis to create an index of marginalization from measures related to education and illiteracy, housing characteristics (like access to electricity and overcrowding), size of population, and income.

²⁵The 2018 National Health and Nutrition Survey (*Encuesta Nacional de Salud y Nutrición* or ENSANUT) did include blood lead measures for children under five. However, this means everyone with a lead measure was

rely on the battery recycling facility location, school location, and wind direction data described in Section 3 to proxy exposure to lead. Given that lead exposure can occur through airborne emissions, soil and dust resuspension, water, food, and take-home contamination on workers' clothing, plant proximity is best read as a proxy for the bundle of pathways rather than for any single one (Clay et al., 2024).

The first measure we use to proxy for lead exposure is the distance between battery recycling facilities and the primary and lower secondary schools. We define students as exposed to lead pollution from ULAB recycling in a given year if they attend a school within five miles of a recycling facility in that year. Figure 2 illustrates this process. In each panel, a battery recycling facility is indicated by a red point, and primary and lower secondary schools by black points. The red circle around the recycling facility has a radius of 5 miles. We define all schools within this red circle as “near” the recycling facility and, therefore, exposed to lead pollution from the facility. Evidence from U.S. battery recycling facilities, where a network of lead pollution monitors allows researchers to measure how far airborne lead pollution travels, shows that airborne lead concentrations drop steeply at monitors several miles from a ULAB facility (Tanaka et al., 2022). Closely related U.S.-based studies identify contamination within a 1.5 to 2-mile radius around airports, where pollution comes from leaded aviation fuel, and lead smelters (Miranda, Anthopolos, & Hastings, 2011; Currie, Lucas, Greenstone, & Walker, 2015), and researchers have used similar distances as a proxy for exposure to study various types of toxic pollutant exposure (e.g., C. L. Persico and Venator (2021)). At the same time, traces of lead have been detected more than 100 miles from lead sources (Widory, Vautour, & Poirier, 2018; McConnell et al., 2018), implying that a tighter cutoff may miss genuine exposure. We therefore adopt a five-mile threshold as our baseline and restrict the control group to schools within 30 miles of a plant.²⁶

born after the regulatory change in 2009. Furthermore, since the data are intended to be representative at the national level, there are very few observations near lead recycling plants. Using confidential location data at INEGI's (Mexico's statistical agency) data lab, we confirmed this is the case.

²⁶We check how sensitive our estimates are to these choices (i.e., 5 and 30 mile cutoffs) with specification variations described in Section 4.2. Overall, the results are consistent across different definitions of near and across different maximum cutoffs used to define control schools.

We can additionally proxy for lead pollution exposure using wind direction. Environmental scientists have shown that environmental lead concentrations, often tested in topsoil, and individual exposure around known lead pollution sites are correlated with prevailing winds (Karali, Stavridis, Loupa, & Rapsomanikis, 2020; Johnston et al., 2019; Zhang et al., 2016; Lidsky & Schneider, 2003). Winds blow the pollution downwind of the source, where it settles in the topsoil. Given this, we can further refine our definition of lead-exposed students by identifying schools that are both near a battery recycling facility, as defined above, and downwind of the prevailing winds.

We define the prevailing wind direction for a plant as the 90-degree sector from which the wind blows the highest percentage of the time. Downwind schools are then considered as those that fall within a 90-degree cone that is directly opposite, plus or minus 180 degrees from the prevailing wind cone. For example, if the wind most frequently blows directly from the North (midpoint at 0°), the baseline downwind cone spans 135 to 225 degrees (45 degrees East of South to 45 degrees West of South). All schools within that cone are classified as downwind. We also test the robustness of this definition by narrowing the downwind cone. This method focuses on lead pollution emitted into the air and dispersed by wind. It ignores other potential pathways of lead exposure that may not vary with wind direction, for example, through transport routes where lead materials are brought in and out of the facilities, via facility staff who can bring lead dust home on their skin, hair, or clothing (Baker Jr et al., 1977), or via waterways if waste is dumped or leeches into surface or ground water. Therefore, our comparison of students who attend downwind versus more upwind schools may result in a lower bound estimate of the effect of lead exposure from the post-2009 increase in battery recycling activity. This is because even students at upwind schools may still be exposed to lead pollution from the facility via non-air pathways.

4.2 Empirical Strategy

4.2.1 Main specifications

We implement a difference-in-differences strategy comparing students enrolled in schools within five miles of a battery recycling facility to those more than five miles away (but within 30 miles) in the same state, before and after the 2009 tightening of environmental standards in the United States. Specifically, we estimate the following equation via Ordinary Least Squares:

$$y_{isegt} = \alpha_{seg} + \delta_t + \gamma_{p(s)t} + \beta(Near_{ist} \times Post_t) + \varepsilon_{isegt} \quad (1)$$

where y_{isegt} represents a test score outcome for student i in school s - session e - grade g in year t . $Near_{ist}$ is a dummy variable that equals one if school s in which student i is enrolled at time t is located within five miles of the nearest lead-recycling plant $p(s)$, and zero if the school s is farther away from it but within 30 miles of the nearest plant. The $Post_t$ indicator takes the value 0 before 2009 and the value 1 in 2009 and thereafter. The terms α_{seg} and δ_t represent school-session-grade and year fixed effects, respectively, yielding a two-way fixed-effects specification that accounts for common shocks over time and time-invariant differences across classrooms. Additionally, in our preferred specification, we include plant-by-year fixed effects, $\gamma_{p(s)t}$.²⁷ These fixed effects account for differential trends in school-session-grades' achievement around each plant, reflecting that schools around, for example, a plant in an urban area may be fundamentally different and evolve differently over time than schools around plants in a more rural area. In the results below, we also test the robustness of our results to varying fixed effects. For example, we can include individual fixed effects that account for any time-invariant difference in student

²⁷In some specifications, we also include state-by-year fixed effects. These absorb annual shocks common to all schools in a state (state economic conditions, governor effects, state-level environmental policy). These are not collinear with the plant-by-year fixed effects: approximately 10 percent of schools in our sample are closest to a plant in a different state, leaving sufficient cross-state variation within plant-years to identify both sets of fixed effects simultaneously.

characteristics and grade g by year t fixed effect that allows test scores trends to vary by grade. Our coefficient of interest is β , which captures the differential effect of increased lead exposure due to the U.S. environmental regulation on test scores at schools within five miles of the closest battery-recycling plant. Standard errors are clustered at the school level, as this is the level at which the lead exposure indicator is defined.

The identification assumption is that, in the absence of the 2009 policy change and conditional on the included fixed effects, students near and far from their closest plant would have exhibited parallel trends in test scores. We examine how plausible this assumption is by running event studies of the form:

$$y_{isegt} = \alpha_{seg} + \delta_t + \gamma_{p(s)t} + \sum_{\tau=2006}^{2007} \beta_{\tau}(Near_{ist} \times \mathbb{1}(t = \tau)) + \sum_{\tau=2009}^{2013} \beta_{\tau}(Near_{ist} \times \mathbb{1}(t = \tau)) + \varepsilon_{isegt} \quad (2)$$

where β_{τ} captures the effect of being close to a battery-recycling plant on students' scores τ periods before or after the US environmental reform. This specification allows us to test for differential trends before 2009, which would suggest that the parallel trends assumption may be violated. We expect all the β_{τ} 's in the pre-2009 years to be indistinguishable from zero, which would be consistent with the parallel trends assumption.

4.2.2 Addressing other identification concerns

The primary threat to our identification is that recycling facilities may affect nearby schools through channels other than lead pollution. One such concern is increased economic activity, such as employment growth, following the post-2009 ramp-up in ULAB recycling activity. These economic effects are likely stronger in areas closer to plants. While we do not observe employment at the facilities, we do observe that school enrollment within five miles of a recycling facility increases relative to enrollment in more distant schools (see Figure 4). This pattern is consistent with families relocating for job opportunities, which could alter student composition in ways that bias our estimates. We address this concern in four ways.

First, using the individual-level data, we also estimate a specification with student fixed effects. This approach mitigates concerns that unobserved, time-invariant differences across students — such as ability or family background — might bias our results if students selectively sort in and out of schools near recycling facilities. The estimating equation is:

$$y_{isegt} = \theta_i + \delta_t + \gamma_{p(s)t} + \beta(Near_{ist} \times Post_t) + \varepsilon_{isegt}, \quad (3)$$

where θ_i denotes individual (student) fixed effects.

Second, to directly address concerns that potential migration following the 2009 U.S. regulatory shift could bias our estimates, we redefine exposure based on students' 2008 location, one year prior to the policy change. This approach is analogous to an “intent-to-treat” design, since it fixes treatment assignment at the pre-policy location. Specifically, we estimate a version of equation 1 defining the treatment indicator by a student's 2008 school location:

$$y_{isegt} = \alpha_{seg} + \delta_t + \gamma_{p(s)t} + \beta(Near_{i,2008} \times Post_t) + \varepsilon_{isegt} \quad (4)$$

Third, we restrict the sample to students who have the same value of $Near_{ist}$ across time (i.e., students who neither move from a near-plant school to a far one nor vice versa), thereby side-stepping sorting issues.

Finally, we use wind direction to differentiate between upwind and downwind schools. Specifically, we interact the $Near_{ist}$ indicator in equation 1 with a downwind dummy (and add the necessary two-way interactions), as described in Section 4.1.²⁸ The environmental science literature has shown that airborne lead moves with prevailing winds, but we do not expect the economic effects of a post-2009 ramp-up at a recycling facility to vary with

²⁸Specifically, we estimate a $y_{isegt} = \alpha_{seg} + \delta_t + \gamma_{p(s)t} + \beta_1(Near_{ist} \times Post_t) + \beta_2(Down_{ist} \times Post_t) + \beta_3(Near_{ist} \times Down_{ist} \times Post_t) + \varepsilon_{isegt}$, where $Down_{ist}$ equals one if school s lies within a 90° cone centered on the direction opposite the plant's prevailing wind. The coefficient of interest is β_3 .

the orientation of prevailing winds. In this specification, the exposed group is made up of those schools where students are exposed to both the increased pollution and economic activity, whereas the non-exposed group, including nearby upwind schools, is made up of students who are exposed to the change in economic activity but not, or at least relatively less, exposed to the lead pollution.

4.3 Results

4.3.1 Estimates of the effect of post-2009 used lead-acid battery recycling on test scores

Following our main specification (equation 1), the 2009 U.S.-induced increase in battery recycling activity reduced standardized math test scores for students enrolled in schools within five miles of a used lead-acid battery recycling facility, relative to more distant schools in the same state (see Table 1, Panel A, Column 1). The point estimate in our preferred specification is -0.027σ (SE 0.006). The effect on Spanish scores is also negative (-0.019σ , SE 0.0054), but of smaller magnitude (Panel B).

The effects are robust to including different sets of fixed effects in Columns 2–5, including grade-by-year, session-by-year, state-by-year, and plant-by-year. The point estimates are relatively stable across specifications; adding plant-by-year fixed effects yields qualitatively similar results. These estimates provide evidence that the negative effects are not driven by differential time trends in school-session-grade achievement around each plant.

A causal interpretation requires that, absent the 2009 increase in recycling activity, test scores in schools near ULAB facilities would have evolved parallel to those farther away. Event study estimates (see Figure 3) show no detectable pre-2009 differences between the test scores in schools within five miles of a facility and those farther away. This lack of pre-trends is consistent with the parallel trends assumption, lending support to our causal interpretation of the coefficients in Table 1.

In Appendix Figures A1 and A2, we assess sensitivity to the distance-based treatment definition. The five-mile cutoff used in our baseline is consistent with evidence that

airborne lead concentrations drop steeply several miles from a ULAB facility (Tanaka et al., 2022). At the same time, studies have detected traces of lead more than 100 miles from emission sources (Widory et al., 2018; McConnell et al., 2018), implying that our baseline estimates may represent lower bounds, since some students classified as unexposed could still be affected. Broadly, our results are robust to alternative definitions of proximity. First, holding the five-mile treatment definition fixed, we re-estimate Equation 1 while varying the maximum distance allowed for control schools from 6 to 50 miles. The estimate is near zero when the control group is restricted to schools within five miles —plausibly because control schools are exposed as well — and becomes more negative before stabilizing at -0.04σ once the control group extends beyond roughly 20 miles (Figure A1). Second, we replace the binary $Near_{ist}$ indicator with a full set of distance-ring dummies interacted with $Post_t$ (Figure A2). The estimated effect is largest for schools closest to a plant and attenuates toward zero with distance, consistent with a dose-response relationship in which proximity to the facility, rather than a common local shock, drives the result.

4.3.2 Robustness checks

The main specification compares students at schools within five miles of a recycling plant to students at schools farther away in the same state, before and after 2009. Two concerns threaten this comparison. As mentioned above, the post-2009 ramp-up in battery recycling may have increased labor demand and altered local economic activity, with stronger effects near the plants. Furthermore, pollution and economic changes may have induced selective migration into or out of near-plant schools, thereby altering student composition over time. Figure 4 shows that enrollment at near-plant schools grows relative to more distant schools, so the concern is not dismissible on descriptive grounds. We run four robustness checks, reported in order of increasing restrictiveness.

First, we absorb all time-invariant student characteristics by including individual (student) fixed effects, following equation 3 (Table 2). The point estimates are larger in absolute value than the baseline: -0.0214σ (SE 0.00654) to -0.0315σ (SE 0.00695) for math across the three

specifications, and -0.0153σ (SE 0.00549) to $-0.01 - 0.018\sigma$ (SE 0.00553) for Spanish. This rules out the hypothesis that the baseline estimate is driven by time-invariant compositional differences in who attends near-plant schools.

Second, we freeze each student's treatment status at their last pre-2009 observation and assign them that value for the remainder of the panel (Table 3 and Figure A3). This intent-to-treat specification rules out any post-2009 endogenous move from reclassifying a student. The preferred-specification math coefficient is -0.0163σ (SE 0.00494); the Spanish coefficient is -0.0138σ (SE 0.00423).

Third, we restrict the sample to students whose *Near* indicator is constant over the panel — students who neither move from a near-plant school to a far one nor vice versa (Table 4 and Figure A4). The math coefficient in the preferred specification is -0.0257σ (SE 0.00653) and the Spanish coefficient is -0.0191σ (SE 0.00566).

Fourth, we compare schools downwind (within a 90° cone opposite the plant's prevailing wind) to upwind schools, both within five miles of a plant, before and after 2009 (Table 5). This triple-difference holds any non-wind-correlated local-economy effects constant. The triple-difference coefficient is -0.0698σ (SE 0.0175) for math and -0.060σ (SE 0.0152) for Spanish. The results are similar if we use a 30° cone, albeit less precise and of smaller magnitude (see Table A1).

Taken together, these four exercises all reject the hypothesis that the main-specification estimates are driven by selective sorting or by correlated local-economy effects.

4.3.3 Heterogeneous effects

The event-study results presented earlier (see Figure 3) not only support the parallel-trends assumption but also shed light on how the effects of the post-2009 increase in recycling activity change over time. The negative estimated effect of the battery recycling plants worsens over time for both math and Spanish test scores. One explanation is that the battery recycling industry continued to experience growth after 2009. This is supported

by Tanaka et al. (2022), who show that exports of used lead-acid batteries from the U.S. to Mexico increased in 2009 and continued to grow in the following years.

A second explanation is that age at first exposure matters. The younger a child is, the weaker the blood-brain barrier, allowing ingested lead to more easily reach the brain (Saunders, Liddelow, & Dziegielewska, 2012), and younger children are more likely to ingest lead from topsoil while playing outside (Rees & Fuller, 2020). If effects are larger for children first exposed at younger ages, the estimated effect would grow as later ENLACE cohorts increasingly comprise students whose early childhood fell entirely after 2009. We cannot cleanly separate this channel from continued industry growth, since both predict a worsening trend.

The dynamic effects discussed above examine how effects may vary with the amount and severity of lead exposure. We are also interested in exploring how family, community, and institutional factors influence the relationship between lead exposure and learning. To test for heterogeneity along these lines, we use the locality- and municipality-level data described in Section 3.5. We examine how our estimated effects vary by pre-2009 characteristics, including adult educational attainment, infrastructure access, malnutrition rates, formal employment status, regional income inequality, and marginalization. To do this, we first define indicator variables equal to 1 if a classroom is in a community with an above-median value of each characteristic.²⁹ To estimate if effect sizes vary for classrooms in the above versus below median groups for a given characteristic, we fully interact the above indicator with the specification in equation 1.

The results of these analyses are presented in Table 6. For ease of interpretation, we estimate equation 1 separately for classrooms above and below the median. For example, Panel A, Column 1 shows that the post-2009 increase in recycling negatively affected

²⁹The median we consider is the 50th percentile of each measure among all classrooms within a five-mile radius of a battery recycling facility. We use this subsample median, rather than the nationwide one, because almost all schools near recycling facilities are relatively disadvantaged. Using the within-subsample median keeps the treated group balanced and focuses the heterogeneity analysis on differences conditional on exposure.

math test scores by -0.051σ (SE 0.010) for students attending schools in relatively less-educated communities, measured by the average years of schooling completed by adults in the community. We estimate a small, statistically insignificant effect (0.003σ) in schools with above median educational attainment.

In Panel A of Table 6, we consider characteristics that are signs of more developed, wealthier, or generally better off communities. In each case, we find that the negative effects of lead are stronger in communities with relatively low values of these characteristics. The gaps are largest for educational attainment and formal employment, proxied here by the share of the community covered by social security. In communities with lower-quality infrastructure, as measured by the percentage of households connected to a sewer, effects are again strong relative to higher-access communities. In Panel B, we observe the opposite pattern, with the measures we consider there taking higher values in worse-off communities. In each case, the effects of lead pollution are larger in communities above the median of these characteristics. The clearest split is on the marginalization index: -0.052σ above-median versus 0.003σ below-median. Taken together, these results tell a consistent story: the negative learning effects of pollution exposure are concentrated in communities that were already worse off before the 2009 increase in recycling activities.³⁰

5 Discussion

How can we contextualize the magnitude of the effects that we estimate? One way to do this is to compare our estimates with the effects of policies that aim to increase educational attainment. A recent review of education interventions with impact evaluations in low- and middle-income countries found that, at the median, these interventions increase learning outcomes by 0.10 standard deviations. These interventions include programs aimed at both increasing access to education and in-class learning. An education intervention with this effect size could reverse the effects of lead pollution in our setting (Evans & Yuan, 2022).

³⁰We also show that the students in schools with below median scores before 2009 are also more affected by lead pollution (see Table A2).

From Mexico, a study examining the effect of a program that increased the time students spend in the classroom found that full-time school increased test scores by 0.11 standard deviations (Cabrera-Hernández, 2020). The effects of this program were concentrated among poorer students, who may be similar to those in our study who live in marginalized communities near lead acid battery recycling facilities.

Our estimated effect sizes are largely consistent with the existing literature on the relationship between pollution and test scores. Compared to studies that consider air pollution, usually defined as PM10 or PM2.5, our estimated effects are similar. Studies that estimate the effects of day-of-test particulate matter pollution on test score outcomes find a negative effect of 0.03 to 0.05 standard deviations, with effects being stronger for more analytical subjects like mathematics (Amanzadeh, Vesal, & Ardestani, 2020; Ebenstein, Lavy, & Roth, 2016). Looking at the effects of longer-term, at-school exposure to air pollution from vehicle traffic, Heissel, Persico, and Simon (2022) find that attending schools downwind of a major highway is associated with a 0.04 standard deviation decrease in test scores. In a U.S.-based study linking lead exposure and education, which used blood lead level (BLL) testing data to measure lead exposure precisely, the authors found that interventions to decrease BLLs resulted in a 0.12 standard deviation increase in educational performance, compared to those students who were not eligible for the intervention (Billings & Schnepel, 2018). This benchmark is relevant because it suggests that targeted lead abatement interventions in our setting could plausibly reverse the losses we document; a point we return to in the cost-benefit calculation below.

We translate our estimates into monetary value by calculating the impact of lower test scores on students' lifetime earnings. We focus on the cohort of students who were in 3rd grade in 2009 and attended schools within five miles of a ULAB recycling facility. At that time, the present value of expected lifetime earnings for a 3rd grader was approximately USD 240,750, using an annual average wage of USD 18,806 in 2016 dollars

(OECD, 2024), a 5% discount rate³¹, and Mexico’s average wage growth of 0.3% between 1991 and 2022 (OECD, 2023).³² Evidence from Mexico links a one standard deviation change in grade 12 test scores to a 6% change in earnings (De Hoyos, Estrada, & Vargas, 2021), while estimates from the U.S. and Sweden suggest between 6 and 15% (Chetty et al., 2014; Lindqvist & Vestman, 2011). Applying our most conservative estimate of the effect on math scores, -0.016σ (main specification, Table 1), we estimate that the post-2009 growth in ULAB recycling reduced lifetime earnings of each 3rd grader by between 230 and 580 USD.³³ Given that 96,711 3rd graders studied within five miles of a ULAB recycling plant in 2009, the aggregated cost to them in terms of lifetime earnings is on the order of USD 22.3–55.9 million per cohort.

Each year, a new cohort of children is born into communities surrounding ULAB recycling plants, inheriting these same burdens. Thus, we consider the cost above to be the annual damage from pollution resulting from the post-2009 increase in ULAB recycling. This estimate likely understates the true cost, since lead pollution exposure is particularly harmful to young children, and the resulting losses in lifetime earnings per cohort extend beyond our observation window. Furthermore, this estimate excludes other health-related effects of lead pollution exposure, including adverse effects on cardiovascular and renal health.

Estimates of the costs of reducing lead pollution emissions from ULAB recycling activities come from Burr, Lazzari, and Greene (2011), who estimated that the cost of bringing the 14 ULAB recycling plants in the U.S. into attainment with the updated 2009 air quality standards was USD 10.33 million per year (2016 dollars). Scaling that estimate proportionally to the 26 facilities operating in Mexico yields a benchmark abatement cost of roughly USD 19.2 million per year. Comparing this benchmark to the annual

³¹We apply the 5% discount rate following Chetty, Friedman, and Rockoff (2014).

³²The present value assumes earnings from age 18 to 65, discounted to the date the cohort was in 3rd grade.

³³We use the range of test score to future earning relationship estimates here given that the Mexico estimate is based only on secondary school graduates and considers their earnings at ages 18-20. This likely underestimates the relationship because it ignores students who have not completed secondary school, a large share in Mexico, and are likely to earn significantly less.

lifetime-earnings loss we estimate (USD 22.3–55.9 million per cohort) suggests the costs of lead pollution abatement from ULAB recycling facilities are likely outweighed by the potential gains from increased learning and incomes.

6 Conclusion

In this paper, we provide causal evidence on the educational cost of a shift in lead-acid battery recycling from the U.S. to Mexico, exploiting a 2009 U.S. regulatory change that caused recycling operations to move across borders. Using census-based standardized test data (ENLACE) and a quasi-experimental research design, we show that exposure to the post-2009 expansion in battery recycling reduced math scores by 0.015 to 0.027 standard deviations in our preferred specification, with smaller but still negative effects in Spanish. Wind-based triple-difference estimates that net out co-located local-economy effects yield larger magnitudes, on the order of -0.07σ for math and -0.06σ for Spanish. These impacts persist for at least four years after first exposure, grow over time, and are concentrated in more marginalized communities. The results are robust to individual-level fixed effects, to intent-to-treat with treatment frozen in 2008, to restricting the sample to never-movers, and to the wind-based triple-difference — and are therefore unlikely to be driven by selective sorting or by other local changes accompanying the post-2009 growth in Mexico’s battery recycling industry.

Our findings also have important implications for environmental, trade, and education policy. First, our analyses build on [Tanaka et al. \(2022\)](#) by further quantifying the magnitude of the environmental externality arising from ULAB recycling shifting from the U.S. to Mexico, and underscoring the need for cross-border coordination of environmental policy. From an educational perspective, because we study the impact of lead exposure on academic outcomes during school years (rather than during early childhood or gestation), our results point to the importance of investing in compensatory measures

to protect students from exposure to school-related pollution.³⁴ Since the educational damage is concentrated in already marginalized communities, those with lower adult education, weaker infrastructure, and higher malnutrition rates, pollution displacement of this kind likely widens existing inequalities, and cost estimates extrapolated from rich-country settings may understate true LMIC welfare losses.

Despite decades of research showing the negative health effects of lead pollution, global exposure rates remain high (Rees & Fuller, 2020). The vast majority of the costs associated with lead pollution exposure fall on children living in low- and middle-income countries (Larsen & Sánchez-Triana, 2023; Crawford et al., 2025). Rich countries have, and continue to, work to clean lead out of their environments, but, as in the setting we studied here, these environmental policies have at times worsened the problem in LMICs by shifting pollution sources to areas with weaker or less-enforced regulatory standards. There is still much work to be done to combat this environmental challenge and mitigate the global health and economic costs associated with lead pollution exposure.

³⁴Although our analysis captures exposures beginning in kindergarten, it does not extend to prenatal or infant periods, which remain outside the scope of our data.

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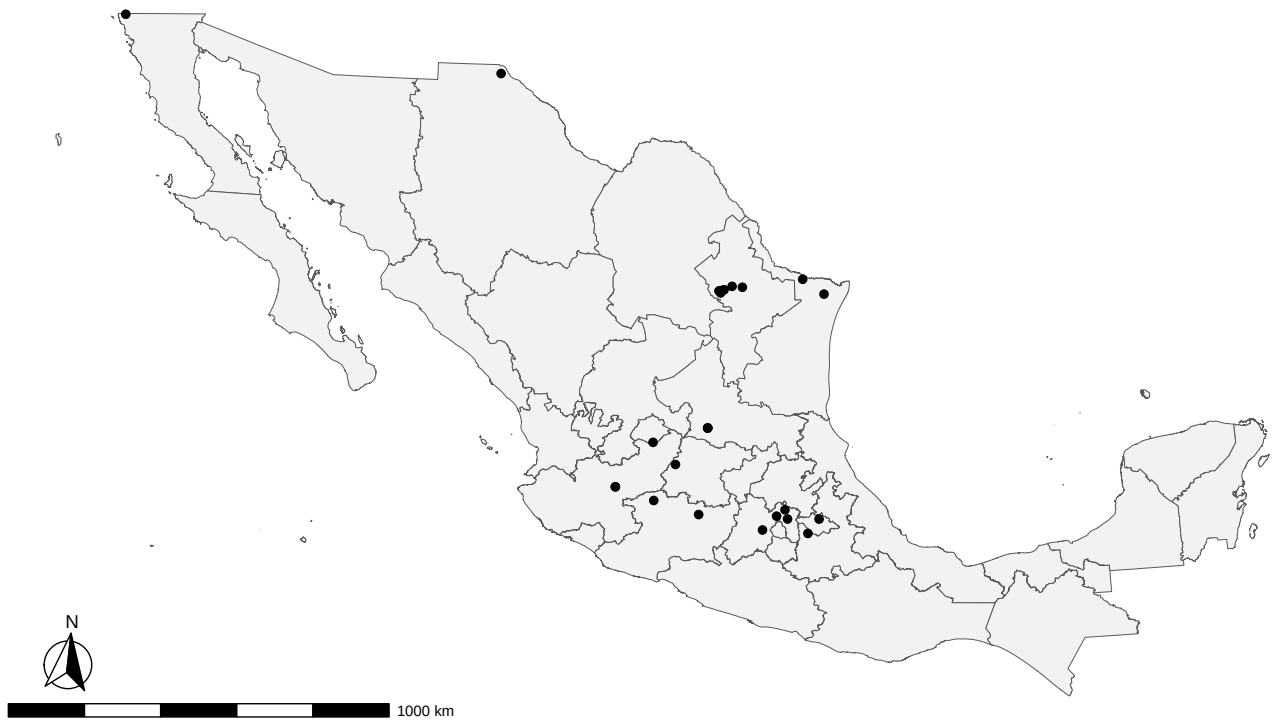
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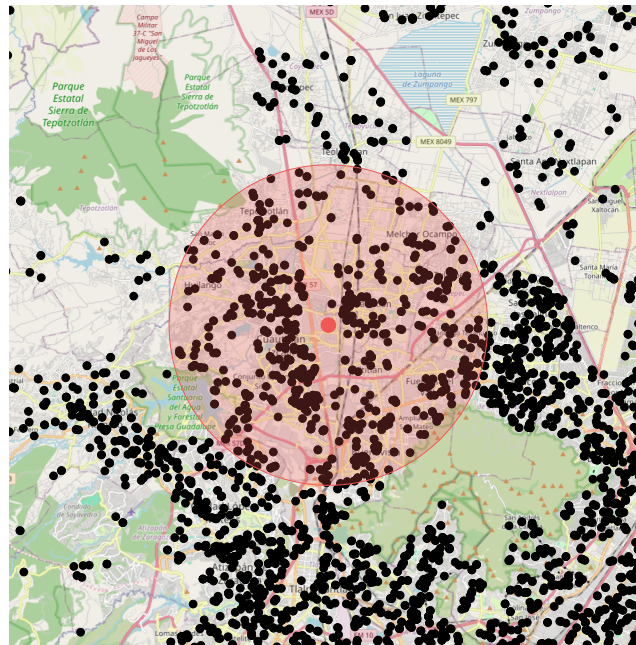
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Figure 1: Used lead-acid battery Recycling Facilities in Mexico

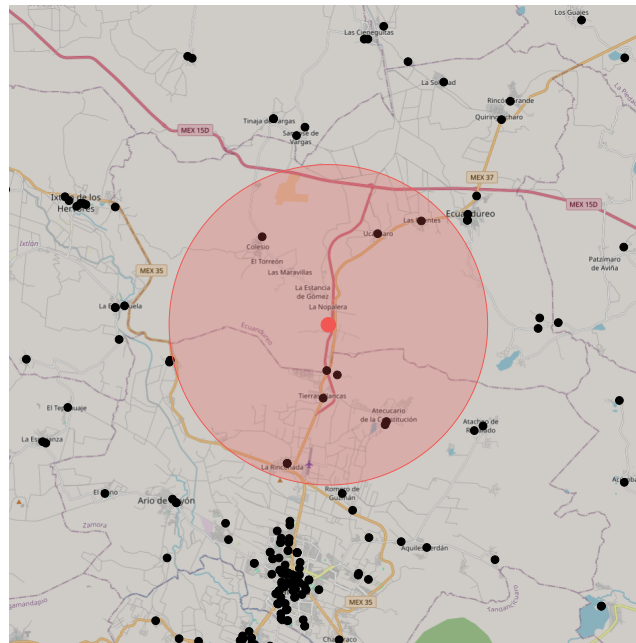


Note: The 26 facility locations plotted here as black points were compiled by [Tanaka et al. \(2022\)](#) from the CEC and SEMARNAT. Some facilities are very close to one another, making them difficult to distinguish at this scale.

Figure 2: Identifying lead-exposed schools



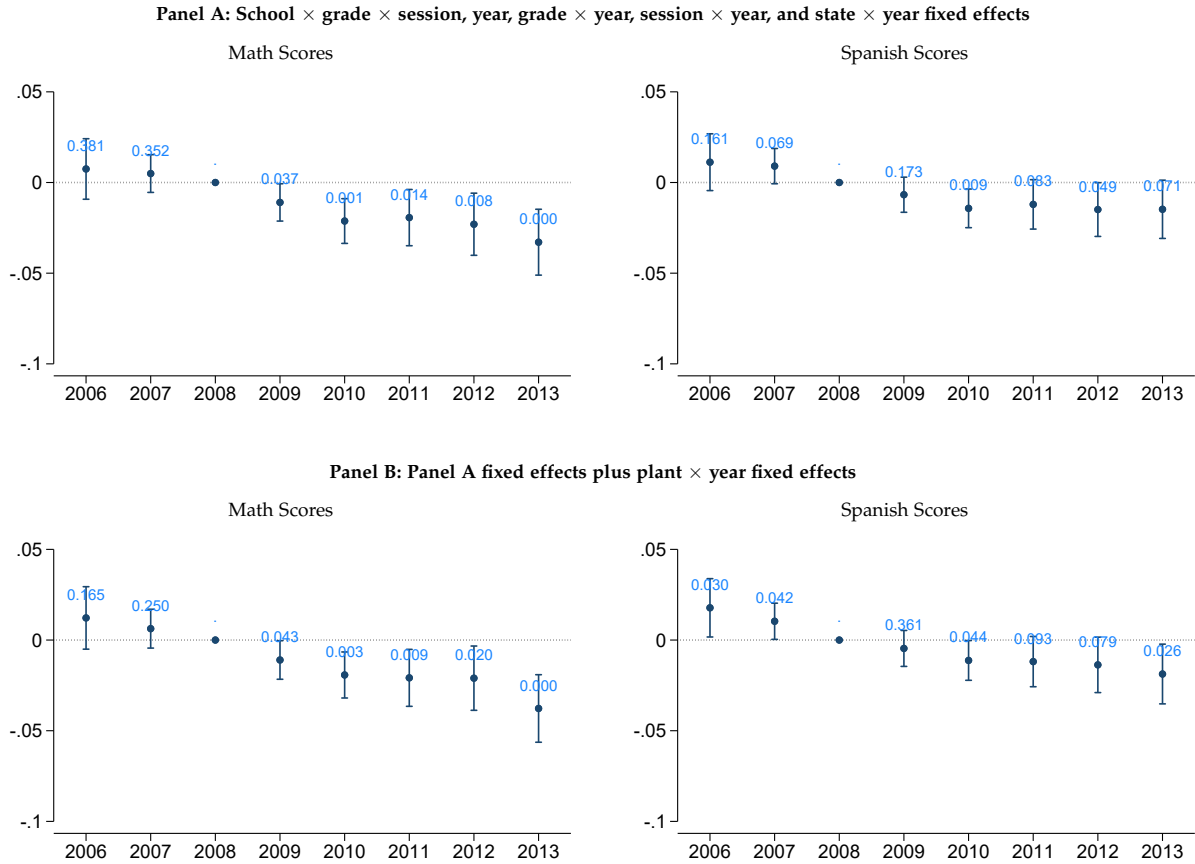
(a) Used lead-acid battery recycling facility and nearby schools in Greater Mexico City



(b) Used lead-acid battery recycling facility and nearby schools in a rural part of Michoacán State

Note: Used lead-acid battery recycling facilities are indicated by a red point. Black points indicate schools at which the ENLACE test was conducted between 2006 and 2013. The red circle represents a five-mile radius around the recycling facility.

Figure 3: Effects of being close to a lead-acid battery recycling plant on scores, by year



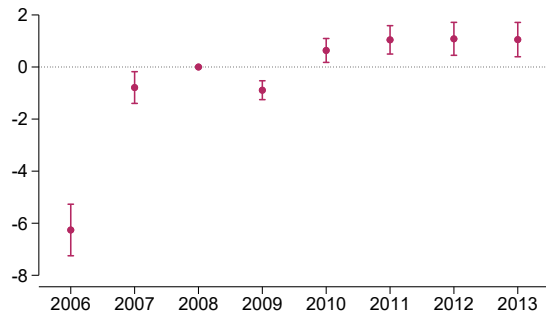
Notes: Figure displays the coefficients and 95% confidence intervals from the event-study of being within five miles of a recycling plant when looking at standardized math and Spanish scores as outcomes. The estimating sample is students in grades 3, 4, 5, 6, and 9 attending schools within 30 miles of the nearest lead-acid battery recycling plant. Panel A includes school $\times$ grade $\times$ session, year, grade $\times$ year, session $\times$ year, and state $\times$ year fixed effects; Panel B adds plant $\times$ year fixed effects. These coefficients correspond to the β_τ parameters in equation 2. Standard errors are clustered at the school level. The reference period is 2008, the year before the U.S. EPA tightened its lead air quality standard.

Table 1: Effect of increased battery recycling activity on students attending school close to a recycling facility

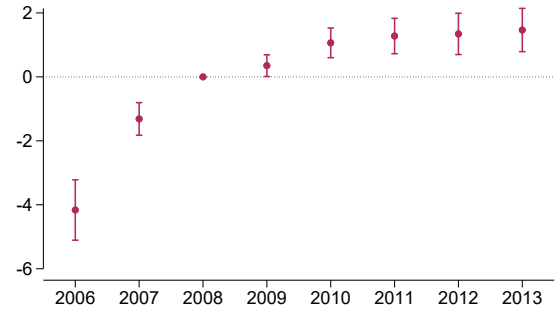
	(1)	(2)	(3)	(4)	(5)
Panel A: Math					
Near Plant $\times$ Post 2009	-0.0155*** (0.00599)	-0.0168*** (0.00595)	-0.0248*** (0.00609)	-0.0266*** (0.00628)	-0.0239*** (0.00633)
Num. of students					-0.00135*** (0.0000642)
Constant	0.0367*** (0.000590)	0.0369*** (0.000586)	0.0376*** (0.000600)	0.0378*** (0.000619)	0.143*** (0.00504)
Observations	23,907,556	23,907,556	23,907,556	23,907,556	23,907,556
R^2	0.136	0.136	0.138	0.138	0.139
Number of schools	26,039	26,039	26,039	26,039	26,039
Panel B: Spanish					
Near Plant $\times$ Post 2009	-0.00695 (0.00519)	-0.00766 (0.00517)	-0.0181*** (0.00525)	-0.0193*** (0.00540)	-0.0174*** (0.00542)
Num. of students					-0.000946*** (0.0000553)
Constant	0.0666*** (0.000512)	0.0667*** (0.000510)	0.0677*** (0.000518)	0.0678*** (0.000532)	0.142*** (0.00435)
Observations	23,895,765	23,895,765	23,895,765	23,895,765	23,895,765
R^2	0.149	0.149	0.151	0.151	0.151
Number of schools	26,039	26,039	26,039	26,039	26,039
<i>Fixed effects:</i>					
School $\times$ Grade $\times$ Session	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes
Grade $\times$ Year	No	Yes	Yes	Yes	Yes
Session $\times$ Year	No	Yes	Yes	Yes	Yes
State $\times$ Year	No	No	Yes	Yes	Yes
Plant $\times$ Year	No	No	No	Yes	Yes

Note: The unit of observation is the student-year. Near is a dummy variable that equals 1 if the student attends a school s located within 5 miles of the nearest lead-recycling plant $p(s)$, and 0 if the school is farther from the nearest plant but within 30 miles. The $Post_t$ indicator takes the value 1 starting in 2009. The sample includes all students in grades 3, 4, 5, 6, and 9 attending schools in states that contain at least one lead-acid battery recycling plant. Outcomes are students' standardized test scores, standardized at the grade-year level. Panel A reports estimates with math scores as the outcome; Panel B reports estimates with Spanish test scores as the outcome. Both panels share the same five-column specification, with the fixed-effects ladder indicated in the bottom rows: school $\times$ grade $\times$ session and year fixed effects enter in all columns; column 2 adds grade $\times$ year and session $\times$ year fixed effects; column 3 adds state $\times$ year fixed effects; column 4 adds plant $\times$ year fixed effects. Column 5 adds the number of students enrolled in the school-session-grade as a control. Standard errors (in parentheses) are clustered at the school level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

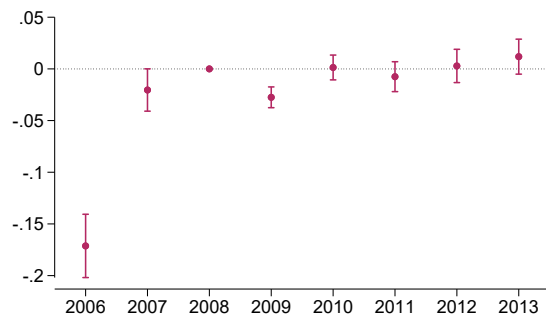
Figure 4: Event study for number of students



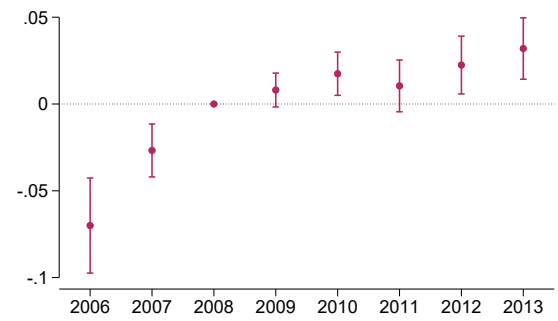
(a) Number of students (basic fixed effects)



(b) Number of students (extended fixed effects)



(c) Log number of students (basic fixed effects)



(d) Log number of students (extended fixed effects)

Note: This figure plots the results of four event studies, estimated for the number of students (top panel) and the log number of students (bottom panel). The left figures include only grade-session-school and year fixed effects, while the right figures include grade-by-year, session-by-year, state-by-year, and plant-by-year fixed effects.

Table 2: Effect of increased battery recycling activity on students attending school close to a recycling facility, individual fixed-effects specification

	Math			Spanish		
	(1)	(2)	(3)	(4)	(5)	(6)
Near Plant $\times$ Post 2009	-0.0214*** (0.00654)	-0.0272*** (0.00688)	-0.0315*** (0.00695)	-0.0174*** (0.00539)	-0.0153*** (0.00549)	-0.0175*** (0.00553)
Constant	0.0418*** (0.000866)	0.0424*** (0.000878)	0.0428*** (0.000885)	0.0697*** (0.000672)	0.0695*** (0.000665)	0.0697*** (0.000669)
Individual	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes
Grade	Yes	Yes	Yes	Yes	Yes	Yes
Grade $\times$ Year	Yes	Yes	Yes	Yes	Yes	Yes
Municipality	No	Yes	Yes	No	Yes	Yes
State $\times$ Year	No	Yes	Yes	No	Yes	Yes
Plant $\times$ Year	No	No	Yes	No	No	Yes
Observations	20,298,752	20,298,751	20,298,751	20,286,333	20,286,332	20,286,332
R^2	0.732	0.734	0.734	0.745	0.746	0.747
Number of schools	25,670	25,669	25,669	25,670	25,669	25,669

Note: The unit of observation is the student-year. The specification includes individual (student) fixed effects, which absorb all time-invariant student characteristics. Near is a dummy variable that equals 1 if the student attends a school s located within 5 miles of the nearest lead-recycling plant $p(s)$, and 0 if the school is farther from the nearest plant but within 30 miles. The $Post_t$ indicator takes the value 1 starting in 2009. The sample includes all students in grades 3, 4, 5, 6, and 9 attending schools in states that contain at least one lead-acid battery recycling plant. Outcomes are students' standardized test scores, standardized at the grade-year level. Columns 1–3 report estimates with math scores as the outcome; columns 4–6 report estimates with Spanish test scores as the outcome. The two outcomes share the same fixed-effects ladder, indicated in the bottom panel: individual, year, grade, and grade $\times$ year fixed effects throughout, with municipality and state $\times$ year fixed effects added in columns 2 (and 5), and plant $\times$ year fixed effects added in columns 3 (and 6). Standard errors (in parentheses) are clustered at the school level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 3: Effect of increased battery recycling activity on students attending school close to a recycling facility, intent-to-treat with treatment frozen at last pre-2009 observation

	Math					Spanish				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Near Plant $\times$ Post 2009	-0.00950* (0.00502)	-0.0106** (0.00502)	-0.0167*** (0.00483)	-0.0163*** (0.00494)	-0.0150*** (0.00494)	-0.00686 (0.00428)	-0.00759* (0.00429)	-0.0152*** (0.00414)	-0.0138*** (0.00423)	-0.0133*** (0.00423)
Num. of students					-0.000916*** (0.0000706)					-0.000370*** (0.0000590)
Constant	0.0517*** (0.000298)	0.0518*** (0.000298)	0.0521*** (0.000287)	0.0521*** (0.000294)	0.129*** (0.00592)	0.0825*** (0.000255)	0.0825*** (0.000255)	0.0830*** (0.000246)	0.0829*** (0.000252)	0.114*** (0.00495)
School $\times$ Grade $\times$ Session	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Grade $\times$ Year	No	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes
Session $\times$ Year	No	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes
State $\times$ Year	No	No	Yes	Yes	Yes	No	No	Yes	Yes	Yes
Plant $\times$ Year	No	No	No	Yes	Yes	No	No	No	Yes	Yes
Observations	13,014,997	13,014,997	13,014,997	13,014,997	13,014,997	13,003,203	13,003,203	13,003,203	13,003,203	13,003,203
R^2	0.166	0.167	0.169	0.169	0.169	0.171	0.172	0.174	0.174	0.174
Number of schools	26,257	26,257	26,257	26,257	26,257	26,258	26,258	26,258	26,258	26,258

Note: The unit of observation is the student-year. Each student's school assignment, treatment status, and nearest plant are frozen at their last observed value before 2009; from 2009 onward, we use these frozen values regardless of whether the student subsequently moves. Students who were never observed before 2009 (or last observed in grade 9 before 2009) are dropped. Near is a dummy variable that equals 1 if the student's last pre-2009 observation reports a school s located within 5 miles of the nearest lead-recycling plant $p(s)$, and 0 if the school is farther from the nearest plant but within 30 miles. The $Post_t$ indicator takes the value 1 starting in 2009. The sample includes all students in grades 3, 4, 5, 6, and 9 attending schools in states that contain at least one lead-acid battery recycling plant. Outcomes are students' standardized test scores, standardized at the grade-year level. Columns 1–5 report estimates with math scores as the outcome; columns 6–10 report estimates with Spanish test scores as the outcome. The two outcomes share the same fixed-effects ladder, indicated in the bottom panel: school $\times$ grade $\times$ session and year fixed effects throughout, with grade $\times$ year, session $\times$ year, state $\times$ year, and plant $\times$ year fixed effects added sequentially in columns 2–4 (and 7–9). Columns 5 and 10 add the number of students enrolled in the school-session-grade as a control. Standard errors (in parentheses) are clustered at the school level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 4: Effect of increased battery recycling activity on students attending school close to a recycling facility, restricting to students who never change treatment status

	Math					Spanish				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Near Plant $\times$ Post 2009	-0.0112* (0.00622)	-0.0127** (0.00617)	-0.0231*** (0.00634)	-0.0257*** (0.00653)	-0.0229*** (0.00657)	-0.00438 (0.00543)	-0.00492 (0.00541)	-0.0174*** (0.00551)	-0.0191*** (0.00566)	-0.0171*** (0.00568)
Num. of students					-0.00137*** (0.0000651)					-0.000960*** (0.0000561)
Constant	0.0355*** (0.000559)	0.0356*** (0.000555)	0.0365*** (0.000569)	0.0368*** (0.000587)	0.143*** (0.00510)	0.0646*** (0.000488)	0.0646*** (0.000486)	0.0657*** (0.000495)	0.0659*** (0.000509)	0.141*** (0.00440)
School $\times$ Grade $\times$ Session	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Grade $\times$ Year	No	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes
Session $\times$ Year	No	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes
State $\times$ Year	No	No	Yes	Yes	Yes	No	No	Yes	Yes	Yes
Plant $\times$ Year	No	No	No	Yes	Yes	No	No	No	Yes	Yes
Observations	23,086,767	23,086,767	23,086,767	23,086,767	23,086,767	23,075,329	23,075,329	23,075,329	23,075,329	23,075,329
R^2	0.137	0.137	0.139	0.139	0.140	0.151	0.151	0.152	0.152	0.153
Number of schools	26,035	26,035	26,035	26,035	26,035	26,035	26,035	26,035	26,035	26,035

Note: The unit of observation is the student-year. The sample is restricted to students who do not change near/far status during the panel (i.e., students who do not move between a school within five miles of a lead-acid battery recycling plant and one more than five miles away). Near is a dummy variable that equals 1 if the student attends a school s located within 5 miles of the nearest lead-recycling plant $p(s)$, and 0 if the school is farther from the nearest plant but within 30 miles. The $Post_t$ indicator takes the value 1 starting in 2009. The sample includes all students in grades 3, 4, 5, 6, and 9 attending schools in states that contain at least one lead-acid battery recycling plant. Outcomes are students' standardized test scores, standardized at the grade-year level. Columns 1–5 report estimates with math scores as the outcome; columns 6–10 report estimates with Spanish test scores as the outcome. The two outcomes share the same fixed-effects ladder, indicated in the bottom panel: school $\times$ grade $\times$ session and year fixed effects throughout, with grade $\times$ year, session $\times$ year, state $\times$ year, and plant $\times$ year fixed effects added sequentially in columns 2–4 (and 7–9). Columns 5 and 10 add the number of students enrolled in the school-session-grade as a control. Standard errors (in parentheses) are clustered at the school level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 5: Effect of attending school close to a battery recycling facility, defining exposure using prevailing wind direction (90° cone)

	(1)	(2)
Panel A: Math scores		
Near $\times$ Downwind (90°) $\times$ Post	-0.0698*** (0.0175)	-0.0703*** (0.0184)
Downwind (90°) $\times$ Post	0.0178** (0.00759)	0.0191** (0.00878)
Near $\times$ Post 2009	0.0304*** (0.00977)	-0.00171 (0.0101)
Observations	17,630,073	17,630,073
R^2	0.136	0.138
Num. of schools	18,258	18,258
Panel B: Spanish scores		
Near $\times$ Downwind (90°) $\times$ Post	-0.0600*** (0.0152)	-0.0580*** (0.0161)
Downwind (90°) $\times$ Post	0.0209*** (0.00641)	0.0173** (0.00757)
Near $\times$ Post 2009	0.0332*** (0.00849)	0.00102 (0.00861)
Observations	17,621,068	17,621,068
R^2	0.149	0.151
Num. of schools	18,258	18,258
Fixed effects:		
School-session-grade	✓	✓
Year	✓	✓
Grade-by-year		✓
Session-by-year		✓
State-by-year		✓
Plant-by-year		✓

Note: The unit of observation is the student-year. “Near $\times$ Downwind (90°) $\times$ Post” is the triple-difference coefficient of interest: it equals one if the student attends a school within five miles of the nearest lead-recycling plant $p(s)$ and downwind of that plant in the post-2009 period. “Downwind” is defined as falling within a 90° cone centered on the direction opposite the plant’s prevailing wind, computed from the Global Wind Atlas. The estimating sample is restricted to students attending schools within 30 miles of plants with a clear prevailing wind direction; plants whose wind rose lacks a single dominant direction are dropped. The $Post_t$ indicator takes the value 1 starting in 2009. Outcomes are students’ standardized test scores, standardized at the grade-year level. Panel A reports estimates with math scores as the outcome; Panel B reports estimates with Spanish test scores. The two outcomes share the same fixed-effects ladder, indicated in the bottom panel: column 1 includes school $\times$ grade $\times$ session and year fixed effects; column 2 additionally includes grade $\times$ year, session $\times$ year, state $\times$ year, and plant $\times$ year fixed effects. Standard errors (in parentheses) are clustered at the school level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

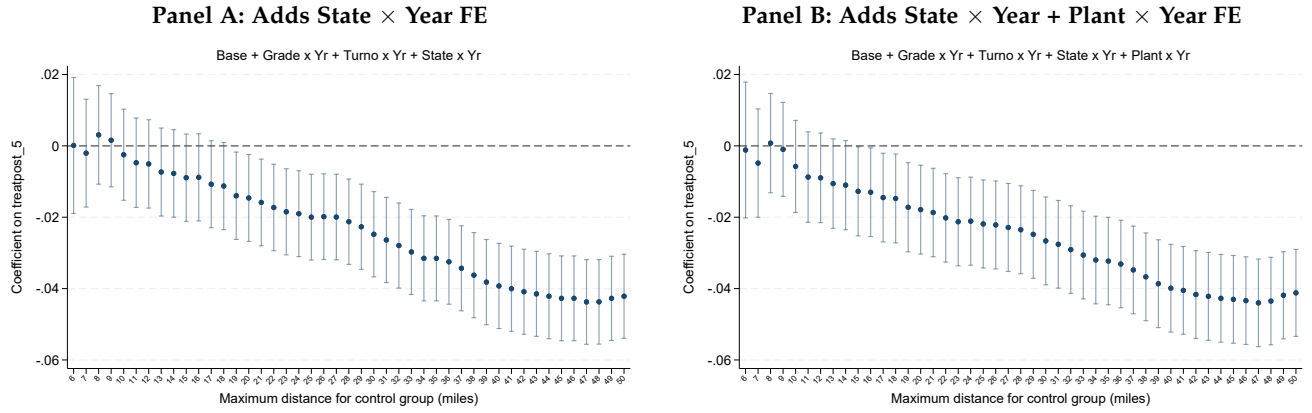
Table 6: Effect of attending school close to a battery recycling facility, by community characteristics

	(1)	(2)	(3)	(4)	(5)	(6)
	Education		Sewer access		Soc. sec. employment	
	Low	High	Low	High	Low	High
<i>Panel A: Education, sewer access, and social-security employment</i>						
Near × Post 2009	-0.0508*** (0.00964)	0.00306 (0.00799)	-0.0367*** (0.00954)	-0.00767 (0.00898)	-0.0185** (0.00901)	-0.00402 (0.00977)
Observations	11,165,332	12,272,492	8,741,016	14,696,909	15,284,914	8,153,011
R ²	0.124	0.140	0.126	0.139	0.129	0.152
Num. of schools	14,369	10,905	11,835	13,440	17,676	7,599
	(1)	(2)	(3)	(4)	(5)	(6)
	Gini		Malnutrition		Marginalization	
	Low	High	Low	High	Low	High
<i>Panel B: Inequality, malnutrition, and marginalization</i>						
Near × Post 2009	-0.00798 (0.00867)	-0.0524*** (0.00971)	0.00633 (0.00928)	-0.0387*** (0.00866)	0.00253 (0.00797)	-0.0517*** (0.00954)
Observations	12,267,513	11,640,043	8,381,043	15,526,513	10,939,027	12,968,529
R ²	0.123	0.153	0.151	0.129	0.146	0.123
Num. of schools	13,042	12,997	8,234	17,805	9,925	16,114
Fixed effects:						
School-session-grade	✓	✓	✓	✓	✓	✓
Year	✓	✓	✓	✓	✓	✓
Grade-by-year	✓	✓	✓	✓	✓	✓
Session-by-year	✓	✓	✓	✓	✓	✓
State-by-year	✓	✓	✓	✓	✓	✓
Plant-by-year	✓	✓	✓	✓	✓	✓

Note: The unit of observation is the student-year. The estimates in this table are produced using the same specification as the estimates presented in Table 1, column 4, with the standardized math test score as the outcome. The specification is run separately for students in communities with above- and below-median values of the given characteristic, with medians calculated from schools within five miles of a battery recycling facility. Panel A reports heterogeneity by educational attainment, sewer access, and the share of working-age adults employed in social-security-covered jobs. Panel B reports heterogeneity by the locality Gini coefficient, the prevalence of child malnutrition, and the CONAPO marginalization index. Standard errors (in parentheses) are clustered at the school level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

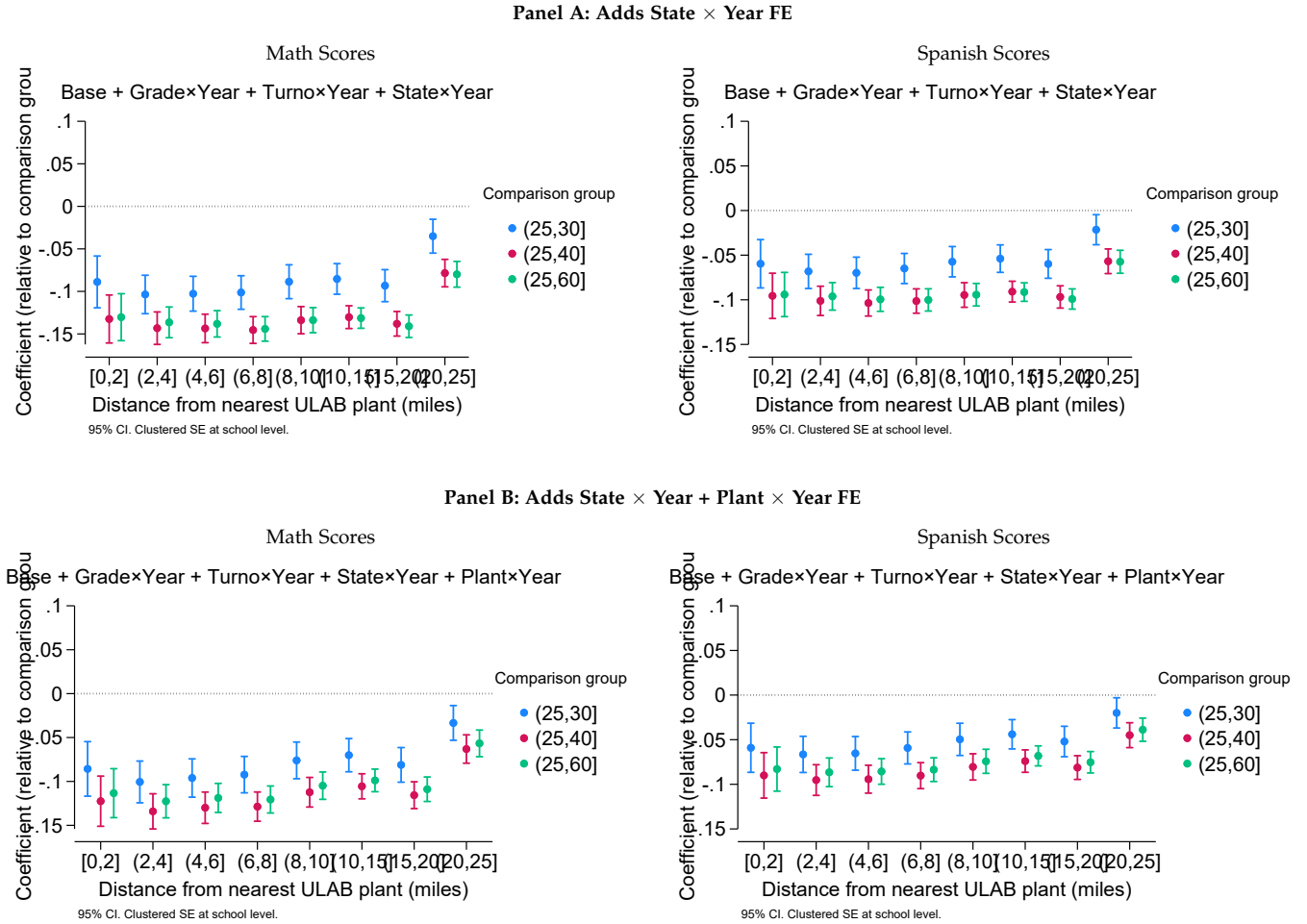
A Additional Table & Figures

Figure A1: Sensitivity of estimated effect to control-group distance cutoff (math)



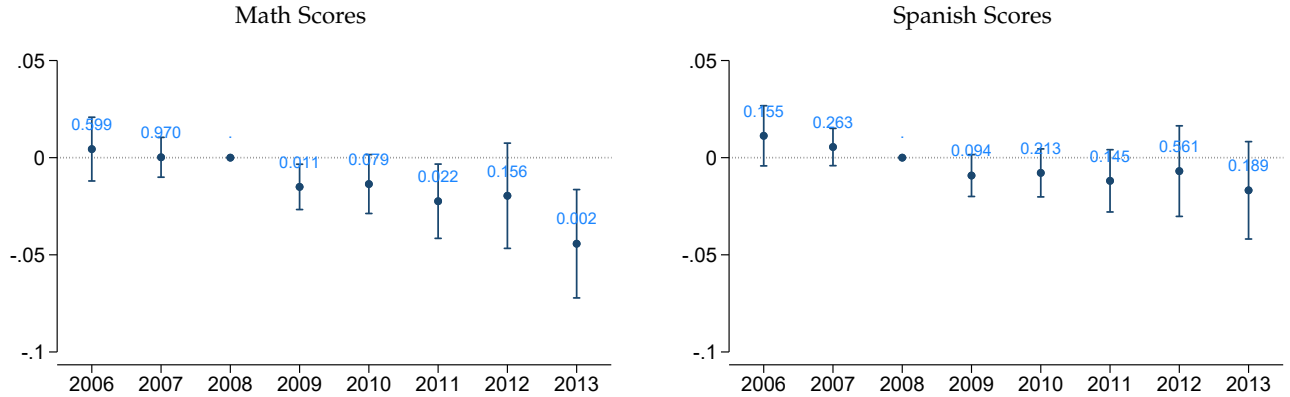
Note: Each marker plots the coefficient on $\text{Near}_s \times \text{Post}_t$ from a separate regression that holds the treatment definition fixed (schools within five miles of the nearest lead-recycling plant) and varies the maximum allowed distance for the control group from 6 to 50 miles. Panel A reports estimates from a specification with school $\times$ grade $\times$ session, year, grade $\times$ year, session $\times$ year, and state $\times$ year fixed effects. Panel B adds plant $\times$ year fixed effects. The sample is restricted to students in grades 3, 4, 5, 6, and 9 attending schools in states that contain at least one lead-acid battery recycling plant. Outcomes are standardized math scores. Vertical bars are 95% confidence intervals constructed from standard errors clustered at the school level.

Figure A2: Donut analysis: estimated effect by distance ring



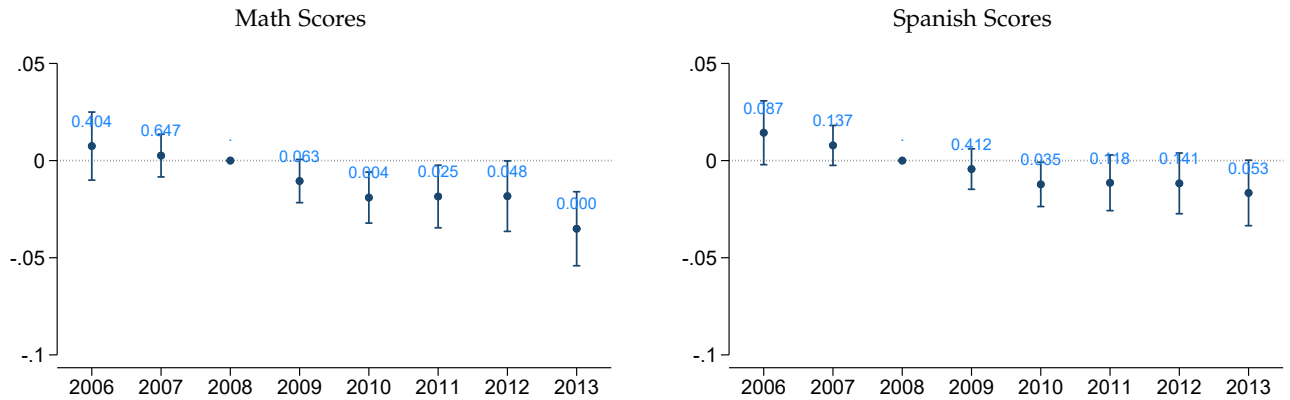
Note: Each marker plots the coefficient on the interaction of a distance-ring indicator with $Post_t$, from a single regression that replaces the binary $Near_s$ measure with a full set of distance-ring dummies. The rings are $[0,2]$, $(2,4]$, $(4,6]$, $(6,8]$, $(8,10]$, $(10,15]$, $(15,20]$, and $(20,25]$ miles, with the $(25, \max]$ ring as the omitted category. Three sample restrictions on the maximum distance are estimated: 30, 40, and 60 miles, so the omitted ring is $(25,30]$, $(25,40]$, or $(25,60]$, respectively. Panel A reports estimates from a specification with school $\times$ grade $\times$ session, year, grade $\times$ year, session $\times$ year, and state $\times$ year fixed effects. Panel B adds plant $\times$ year fixed effects. The left column shows math scores and the right column shows Spanish scores. The sample is restricted to students in grades 3, 4, 5, 6, and 9 attending schools in states that contain at least one lead-acid battery recycling plant. Vertical bars represent 95% confidence intervals based on standard errors clustered at the school level.

Figure A3: Event study of test scores: intent-to-treat (frozen pre-2009 assignment)



Note: The figure plots event-study coefficients on the interaction of year indicators with an indicator for attending a school within five miles of the nearest lead-recycling plant, with 2008 as the omitted base year. Treatment status, the identity of the nearest plant, and the relevant state are frozen at each student's last observation before 2009, so that post-2009 movement does not reclassify students (intent-to-treat). The sample includes all students in grades 3, 4, 5, 6, and 9 attending schools in states that contain at least one lead-acid battery recycling plant within 30 miles at baseline. The specification includes school $\times$ grade $\times$ session, year, grade $\times$ year, session $\times$ year, state $\times$ year, and (frozen) plant $\times$ year fixed effects, and the number of students enrolled in the school-session-grade as a control. The 95% confidence intervals are constructed from standard errors clustered at the school level.

Figure A4: Event study of test scores: students who never change treatment status



Note: The figure plots event-study coefficients on the interaction of year indicators with an indicator for attending a school within five miles of the nearest lead-recycling plant, with 2008 as the omitted base year. The estimating sample is restricted to students whose Near indicator (within five miles of the nearest lead-acid battery recycling plant) does not change across observations, in grades 3, 4, 5, 6, and 9, who attend schools in states containing at least one lead-acid battery recycling plant within 30 miles. The specification includes school $\times$ grade $\times$ session, year, grade $\times$ year, session $\times$ year, state $\times$ year, and plant $\times$ year fixed effects, and the number of students enrolled in the school-session-grade as a control. The 95% confidence intervals are constructed from standard errors clustered at the school level.

Table A1: Effect of attending school close to a battery recycling facility, defining exposure using prevailing wind direction (30° cone)

	(1)	(2)
Panel A: Math scores		
Near $\times$ Downwind (30°) $\times$ Post	-0.0394* (0.0234)	-0.0159 (0.0238)
Downwind (30°) $\times$ Post	0.0184 (0.0124)	-0.00940 (0.0129)
Near $\times$ Post 2009	0.0129 (0.00881)	-0.0206** (0.00915)
Observations	17,630,073	17,630,073
R^2	0.136	0.138
Num. of schools	18,258	18,258
Panel B: Spanish scores		
Near $\times$ Downwind (30°) $\times$ Post	-0.0328 (0.0204)	-0.0137 (0.0208)
Downwind (30°) $\times$ Post	0.0187* (0.0106)	-0.00577 (0.0110)
Near $\times$ Post 2009	0.0190** (0.00766)	-0.0143* (0.00783)
Observations	17,621,068	17,621,068
R^2	0.149	0.151
Num. of schools	18,258	18,258
Fixed effects:		
School-session-grade	✓	✓
Year	✓	✓
Grade-by-year		✓
Session-by-year		✓
State-by-year		✓
Plant-by-year		✓

Note: The unit of observation is the student-year. “Near $\times$ Downwind (30°) $\times$ Post” is the triple-difference coefficient of interest: it equals one if the student attends a school within five miles of the nearest lead-recycling plant $p(s)$ and downwind of that plant in the post-2009 period. “Downwind” is defined as falling within a 30° cone centered on the direction opposite the plant’s prevailing wind, computed from the Global Wind Atlas. The estimating sample is restricted to students attending schools within 30 miles of plants with a clear prevailing wind direction; plants whose wind rose lacks a single dominant direction are dropped. The $Post_t$ indicator takes the value 1 starting in 2009. Outcomes are students’ standardized test scores, standardized at the grade-year level. Panel A reports estimates with math scores as the outcome; Panel B reports estimates with Spanish test scores. The two outcomes share the same fixed-effects ladder, indicated in the bottom panel: column 1 includes school $\times$ grade $\times$ session and year fixed effects; column 2 additionally includes grade $\times$ year, session $\times$ year, state $\times$ year, and plant $\times$ year fixed effects. Standard errors (in parentheses) are clustered at the school level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A2: Effect of attending school close to a battery recycling facility, by baseline (pre-2009) test scores

	(1) Below median	(2) Above median	(3) Interacted
Panel A: Math scores			
Near $\times$ Post 2009	-0.0329*** (0.00737)	-0.0169** (0.00762)	-0.0329*** (0.00737)
Above-median baseline $\times$ Near $\times$ Post 2009			0.0160* (0.00883)
Observations	11,532,622	11,471,253	23,003,875
R^2	0.111	0.144	0.138
Num. of schools	18,054	17,391	22,641
Panel B: Spanish scores			
Near $\times$ Post 2009	-0.0238*** (0.00653)	-0.0130** (0.00650)	-0.0238*** (0.00653)
Above-median baseline $\times$ Near $\times$ Post 2009			0.0108 (0.00772)
Observations	11,525,429	11,465,534	22,990,963
R^2	0.120	0.158	0.150
Num. of schools	18,054	17,391	22,641
Fixed effects:			
School-session-grade	✓	✓	✓
Year	✓	✓	✓
Grade-by-year	✓	✓	✓
Session-by-year	✓	✓	✓
State-by-year	✓	✓	✓
Plant-by-year	✓	✓	✓
All FE interacted with baseline indicator			✓

Note: The unit of observation is the student-year. Baseline test-score status is assigned at the school-session-grade level: a school-session-grade is classified as “Above median” if its average z_{mat} in 2006 is at or above the within-grade median across schools located within five miles of a recycling plant; if 2006 data are missing, the rule falls back first to 2007, then to 2008. Columns 1 and 2 estimate the main specification (Table 1, column 4) separately on the below-median and above-median samples. Column 3 pools the two samples and interacts the indicator for above-median baseline with both $Near_{ist}$ and the full set of fixed effects, so the reported triple interaction tests whether the post-2009 effect differs across baseline groups. The sample includes students in grades 3, 4, 5, 6, and 9 attending schools in states that contain at least one lead-acid battery recycling plant within 30 miles. Outcomes are students’ standardized test scores, standardized at the grade-year level. Panel A reports estimates with math scores as the outcome; Panel B reports estimates with Spanish test scores. The fixed-effects ladder, indicated in the bottom panel, includes school $\times$ grade $\times$ session, year, grade $\times$ year, session $\times$ year, state $\times$ year, and plant $\times$ year; in column 3, each fixed effect is interacted with the baseline indicator. Standard errors (in parentheses) are clustered at the school level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.